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Editorial Office
Risks
MDPI AG, St. Alban-Anlage 66
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Dear Editors,

We wish to submit our manuscript entitled

"Public-Data Causal Multiscale Wavelet Spillover Learning for Stock Index Volatility Forecasting and Risk Early Warning"

for consideration for publication as an original *Article* in *Risks*.

Financial risk context and motivation. Forecasting stock-index volatility and issuing timely high-risk warnings are core tasks in market risk management because they affect portfolio scaling, hedging, margin planning, and stress monitoring. Our manuscript addresses this problem with a reproducible Causal Multiscale Wavelet Spillover Learning (CMWSL) framework that combines HAR persistence features, causal stationary wavelet transform (SWT) summaries, and a cross-index spillover extension based on medium- and long-scale wavelet energy from peer markets. All components are evaluated under a strict walk-forward design so that feature construction, estimation, calibration, and warning generation remain leakage-free.

Key contributions. This study makes four contributions aligned with *Risks*'s scope:

1. **Unified public-data risk pipeline.** We integrate volatility forecasting and high-risk warning inside one causal framework based entirely on public daily market and macro-financial data.
2. **Cross-index wavelet spillover test.** We formalise cross-index wavelet spillover as an incremental-information problem and evaluate it through a nested ablation: HAR $\rightarrow$ Wavelet-LGB $\rightarrow$ Spillover-LGB. Clark–West tests detect significant spillover gains in five of nine index–horizon cells, with the strongest result for the S&P 500 at $h = 10$ (CW $= 4.83$, $p < 0.001$).
3. **Conditional risk diagnostics.** Beyond average forecast rankings, we report conditional diagnostics, including tail, state, rolling-window, and event-based evidence, to show where the multiscale and spillover features matter most.
4. **Reproducible implementation.** The study is fully based on public data, deterministic scripts, and reviewer-ready experiment artifacts so that the main evidence can be audited and rerun without proprietary feeds.

Fit with *Risks*. This manuscript aligns directly with *Risks*'s focus on market risk, volatility forecasting, warning design, and spillover analysis. The paper is finance-specific in its data, prediction tasks, interpretation, and practical implications. It also emphasizes strong baselines, strict walk-forward evaluation, and reproducibility, all of which are important for credible applied risk research.

Originality and ethical statements. This manuscript is original work that has not been published elsewhere and is not currently under consideration at any other journal. All authors have read and approved the final manuscript. No ethical approval was required; only publicly available financial market data (Stooq, FRED) were used.

We are happy to suggest reviewers or provide any additional material at your request. Thank you for your time and consideration.

Sincerely,

Aiping Jiang
(Corresponding Author, on behalf of
all authors)

Author contributions: H.L. — conceptualisation, methodology, software, formal analysis, data curation, visualisation, writing (original draft); Y.S. — validation, writing (review & editing); A.J. — conceptualisation, supervision, writing (review & editing). Funding: This research received no external funding. Conflicts of interest: The authors declare no conflicts of interest.